Which Factors Contribute to the Accuracy and Efficiency of Horse Racing Betting Markets?

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ABSTRACT

This is the abstract.

- A brief introduction to the objectives.
- A brief description of the main work.
- A brief description of the contributions, major findings, results achieved and conclusion.

Nomenclature

BSP	— Betfair Starting Price
CLOB	— Central Limit Order Book
HKJC	— Hong Kong Jockey Club
IP	— In-play (after the off)
PP	— Pre-play (before the off)
WAP	— Weighted Average Price

1. Introduction

Horse racing markets have been academically studied as valuable research targets since at least the 1940s (Griffith, 1949). Racing markets are good to study for a few reasons. For one, they involve a large number of bettors. The bettors have varying levels of expertise ranging from absolutely zero to professional punters. These markets also closely mirror more ''traditional'' financial markets, and so research involving racing can be extrapolated out to the wider context of finance. Betting contracts are similar to options contracts in that they have a set expiry time and a payout determined by an external factor. The facets of horse racing make it an ideal ground to examine the efficient markets model (Snyder, 1978).

As of late, prediction markets have become a hot topic. Services such as Polymarket and Kalshi are offering the opportunity to bet on many world events. There has been research to suggest that prediction markets are more accurate than other forecasting techniques including polls (Berg, 2008). Horse racing markets are also fundamentally prediction markets, and have a long

history of widespread use. As a prediction market, horse races provide good data since they are so regular and the approximate date and time is known before hand. It is easier to determine their accuracy in this context as there are many repeated similar markets, unlike predictions for political events which can be very far apart.

1.1. TEMP

The primary aim of this project is to determine which factors are most significant in influencing the accuracy and efficiency of horse racing markets. There are many different ways to classify horse races including but not limited to time, distance and race type (Hurdles, Chase, Handicap, etc.). Additionally, races can be classified by their market factors such as trade volume or overround.

Accuracy of the market is a measure of how well the listed odds represent the true odds of a horse winning. In a market that was 100% accurate, the odds given would provide a 0% return on all winning bets. To determine the accuracy of the odds, I will be comparing various test statistics such as the Brier score and Log-Loss.

An efficient market is one in which prices are said to ‘‘fully reflect’’ all of the available information and are shown to be *weak-form* efficient when this can be demonstrated using an information set made exclusively of historical price data (Fama, 1970). A systematic betting strategy using historical pricing data that is able to generate a consistent profit demonstrates a lack of *weak-form* efficiency within a market (Hausch, 1981). In order to determine which groups of races are *weak-form* efficient, I shall apply various betting strategy profiles to them in isolation. Some of these strategies will simple and some will be adapted from previous work.

1.2. Motivation

1.3. Scope

1.4. Aims

1.5. Objectives

2. Background

2.1. Bookmaking

2.1.1. Fixed-Price Odds

A fixed-price bookmaker will quote prices (odds) at which they are willing to sell bets at. If the bet pays out, the bettor receives the return originally quoted to them. The bettor is given a guarantee of their winnings before placing the bet. It is important to note that it is unlikely a bookmaker will allow a bettor to ‘‘lay’’ a bet, that is to bet that something *will not* happen. Bookmakers make their money by appropriately setting odds such that in all cases (or at least over a period of time), they will ensure a profit. This is done by changing odds to account for the amount of money bet already (balancing the books) and by providing unfair odds that do not sum to one.

2.1.2. Parimutuel Odds

A tote (otherwise known as parimutuel betting) is a system in which money for all bets is placed into a pool and then distributed to winning bettors. This means that bettors are not given an exact payout quote at the time of purchasing their bet, but they may be given an estimate. Tote operators make money by taking a certain percentage of the pool for themselves, typically called the ‘‘track take’’. These types of bets are typically facilitated by the track operator themselves, such as the HKJC (Hong Kong Jockey Club, 2026).

2.1.3. Betfair Exchange

Betfair is a UK based company which operates a betting exchange service. A betting exchange is different from a typical bookmaker in that instead of the operator quoting odds prices or offering a tote service, it allows users to form bets with one another. Betfair does this using a CLOB, allowing users to either back (long) a bet or lay (short) one. Within the context of a horse race, this means that the win market (where bets are payed out only if the horse places first) features a different asset for every selection (runner) in the race. They can be thought of as largely analogous to options contracts, where the layer agrees to pay the backer a certain sum of money dependant on the outcome of the race.

A unique feature of the Betfair racing market is its "starting price" betting. Bettors who use the starting price option have their bets matched into the market at the off (when the race begins). They are also able to set minimum odds to back or maximum odds to lay. The price determination maximises volume and gives backers and layers the same price. In order to do that, existing positions within the CLOB may be matched against as well. Starting price bets aren’t guaranteed to be matched if the odds limit is at or above the starting price (Betfair,).

2.2. Literature Review

3. Main Results

3.1. Objectives, Specification & Design

3.2. Methodology & Implementation

3.2.1. Data Acquisition

3.2.1.1. Betfair Exchange UK Horse Racing Win Markets

Historical odds pricing data is typically very difficult to find if not impossible from most bookmakers. This was a signifcant factor in choosing to use prices from the Betfair exchange. Betfair provides historical betting data at multiple levels of resolution including a generous free tier. The data provided in this set is structured like exchange data, with multiple entries per race documenting changes in prices over time. For my purposes, this is beyond what I need as I am only interest in the BSP and not the price evolution. I used this dataset at first, but found one better suited to my uses.

Betfair also provides a directory of starting price files and allows users to filter them by location and market type. This dataset contains some useful information not included in the free tier of the standard historical data files such as trade volume and minimum and maximum prices

for each selection. This dataset is also significantly larger, containing data from as far back as 2008, compared to 2016 with the other set.

I produced a Python script to automatically download all of the UK win market files and process them to be used within Pandas. The script saves the data to binary files for quick retrieval. It makes sure to not redownload files it has already obtained so it can easily update the existing database. As well as this, it automatically slows down when rate limited to ensure that all files are downloaded.

The final dataset contains over 170,000 races from 24,000 different events in the UK, starting from September 1st 2008. There are over 1.5 million entries for selections within the races. This data is formatted into three different tables: events, races and, runners.

3.2.1.1.1. Events

Field	Data Type	Notes
`event_id`	`Integer`	Unique identifier provided by Betfair
`location`	`String`	
`event_date`	`Datetime`	

3.2.1.1.2. Races

Field	Data Type	Notes
`race_id`	`Integer`	Unique identifier provided by Betfair
`race_type`	`String`	Indicates the format of the race: handicap, hurdles, chase, etc.
`race_time`	`Datetime`	
`distance_f`	`Integer`	Furlongs are one eigth of a mile long.
`race_no`	`Integer`	Races are given numbers indicating their sequence position within an event.
`event_id`	`Integer`	Relational key.

3.2.1.1.3. Runners

Field	Data Type	Notes
selection_id	Integer	This identifier is unique per horse, not per entry.
selection_name	String	
win_lose	Boolean	Win=True, Lose=False
bsp	Float64	Betfair Starting Price
implied_odds	Float64	1/BSP
ppwap	Float32	Pre-play weighted average price.
morningwap	Float32	Weighted average price of bets before 11AM GMT
ppmax	Float32	Pre-play max price.
ppmin	Float32	Pre-play min price.
ipmax	Float32	In-play max price.
ipmin	Float32	In-play min price.
morningtradedvol	Float32	Trade volume before 11AM
pptradedvol	Float32	Trade volume before the off
iptradedvol	Float32	Trade volume after the off
race_id	Integer	Relational key.

3.2.1.2. Hong Kong Jockey Club

I also downloaded data from the website of the Hong Kong Jockey club in order to compare it against the UK market. This was done by scraping their webpages for historical racing events. Ultimately it proved that this data was not particularly useful, however. The available data I was able to collect only went back to 2022 and the reduced number of racecourses operated by the HKJC meant that the number of races was only around 2000, compared to the 170,000+ of the UK dataset.

3.2.2. Software Development

I decided to use python through Jupyter Lab to write and run my software as it is what I had the most experience with for statistical analysis.

3.2.2.1. Data Structure and Statistics

3.2.2.2. Trading Strategies

3.3. Results, Analysis & Evaluation

3.3.1. Accuracy

3.3.2. Efficiency

4. Legal, Social, Ethical & Professional Issues

- Use of live animals?

- Profit Derived from gambling losers
- Similarity to stock market
- legality of companies betting

5. Conclusion

References

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